HTML

HTML = HyperText Markup Language

HTML is the **skeleton structure** of every webpage you have ever visited. Before YouTube videos play, before Instagram loads, before Google shows results — the browser first receives and parses HTML.

It is NOT a programming language. It is a **markup language** — meaning it uses predefined markup tags to describe content and define its structure and semantic meaning.

HYPERTEXT

"Hyper" means beyond normal.
HyperText = text that can link to other text/pages.
This is what `<a href="..."> ... </a>` does.In the 1990s, this was revolutionary

Markup Language

Markup = applying labels to content. Like a teacher marking exam paper with red pen. You mark up your content with tags to tell the browser what each piece means

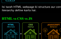
RELATE WITH REAL-WORLD

HTML ek blueprint ki tarah hota hai, jaise koi architect ghar ka structure design karta hai.

- Darwaza kahan hoga
- Khidki kahan hogi
- Room ka layout kya hoga

Isi tarah HTML webpage ki structure aur content hierarchy define karta hai.

HTML vs CSS vs JS



Therefore: HTML provides the structure, CSS handles presentation, and JavaScript adds behavior.

INTERVIEW INSIGHT

01 HTML is not considered a programming language because it cannot perform computations or logical operations.

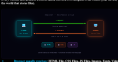
02 HTML stands for HyperText Markup Language.

03 HTML was created by Tim Berners-Lee in 1991.

04 HTML5 is the modern standard of HTML used today.

Client vs Server

Every website interaction is a conversation between two computers: the Client (your device) and the Server (a computer somewhere in the world that stores files).



Browser usually receives: HTML File, CSS Files, JS Files, Images, Fonts, Video/Audio,

Browser Parses HTML

Browser starts reading HTML line-by-line.

Example:

```
<h1>Hello</h1>

<link rel="stylesheet"
  href="style.css">

<script src="app.js"></script>
```

Now browser discovers:

CSS needed
JS needed
images needed

Then browser sends MORE requests.

Additional Requests Happen

Browser again asks server:

Give me style.css
Give me app.js
Give me logo.png

So one webpage may create:

10
50
200+
requests.

Relate with Real-World:

Client = Customer at a restaurant. Server = Kitchen. You (client) place an order (request). The kitchen (server) prepares the food and sends it back (response). You don't know HOW they made it — you just receive the result.

HTML = human body banayega. CSS = style dega i.e. colour kesa hona chaye person ka, colour of dress.
JS = us human body me jaan dalega i.e. ab vo human kisi se interact kar payega.

Why HTML was Created

The 1990s Problem

- Imagine the internet in 1990. There were no links. If you wanted to visit YouTube (hypothetically), you had to type the entire URL manually every single time: https://youtube.com.
- To navigate between pages, there was no way to click a link and go there. You had to memorize every URL. This was incredibly painful.

In short: The web was hard to use, slow to navigate, and not user-friendly.

The Solution: HyperLinks

- Tim Berners-Lee created HTML with the concept of HyperLinks — clickable text that teleports you to another page. This single idea changed the entire internet.
- That's why it's called HYPERText — text that goes BEYOND normal text by being navigable.

Example:

`<a href="https://youtube.com">Go to YouTube</a>`

This one tag replaced the need to memorize URLs!

The Impact: HyperLinks made the web connected, navigable, and scalable. They turned a collection of documents into the World Wide Web.

1989 → WWW idea proposed
1991 → HTML created
1993 → Web browsers became popular

Important: HTML itself did not create the Internet.
HTML made the Internet navigable and user-friendly.

The "WWW" in websites stands for World Wide Web — a system of interconnected documents linked using HyperLinks.

Internet vs WWW

Internet: The physical/global network of computers.

WWW (World Wide Web): A system of interconnected webpages running ON the Internet.

So:

Internet already existed
WWW made it user-friendly and navigable

That's why HTML and hyperlinks became revolutionary.

First Website Ever Created

- Created by Tim Berners-Lee at CERN.
- It was launched in 1991.

CERN is one of the world's largest scientific research organizations.

CERN = European Organization for Nuclear Research

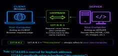
- Link: https://info.cern.ch/hypertext/WWW/TheProject.html

LOCALHOST & PORT NUMBERS

Q)What is Localhost ?

When you create a website, you need a server to serve the files. But do you need a big server in another country? No! Your own computer can act as a server for testing

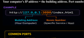
Localhost = your own computer acting as a server. You become both the client AND the server — apna computer hi aapka server ban jata hai


CLIENT (Browser)
LOOPBACK
127.0.0.1
SERVER
Your Computer
Acting as CLIENT
Browser requests page
Request never leaves your computer. It comes back to the same machine.
Your Computer
Acting as SERVER
Serves files (HTML, CSS, JS, images, etc.)
127.0.0.1
127.0.0.1 = "This machine" — always refers to your own computer.

Note: 127.0.0.0/8 is reserved for loopback addresses.
127.0.0.1 specifically refers to localhost ("this computer").

Port Numbers

Your computer's IP address = the building address. Port number = flat/room number inside the building.



COMMON PORTS

PORT	SERVICE	USE
80	HTTP	Websites (unencrypted)
443	HTTPS	Secure websites
3000	Custom / Dev	React / Node.js dev servers
5500	Live Server	VS Code Live Server
8080	HTTP Alt	Backend / Web servers
3306	MySQL	MySQL Database

IMPORTANT NOTES

- ✓ 127.0.0.1 is the loopback address. Requests to this address never leave your machine.
- ✓ localhost is a hostname that usually resolves to 127.0.0.1.
- ✓ You can run multiple servers on the same machine using different ports.
- ✓ Two applications cannot use the same port simultaneously on the same machine.

INTERVIEW INSIGHT

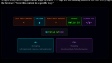
127.0.0.1 and localhost are the same (usually).

- Port numbers range from 0 to 65535.
- Ports 0–1023 are called well-known ports (reserved for standard services).
- Developers use localhost for testing websites, APIs, databases, etc.

HTML Tags, Elements & Syntax

Q/What is a Tag?

A tag is a keyword wrapped inside angle brackets < >. Tags are the building blocks of HTML. Every tag has a purpose and tells the browser: "treat this content in a specific way."



INTERVIEW INSIGHT

① **ELEMENT** — An HTML element usually consists of:
- Opening tag
- Content
- Closing tag

EXAMPLE `<p>Hello</p>` This is a complete HTML element.

② **TAG** — A tag is only the markup syntax part. EXAMPLES `<p>` `</p>` These are tags, not elements.

③ **CASE SENSITIVITY** — HTML tags are case-insensitive, but lowercase is the standard and recommended convention. PREFERRED `<p>` AVOID `<P>`

④ **VOID ELEMENTS** — Void elements do not require closing tags because they cannot contain content. EXAMPLES `<br>` `<hr>` `<img>` `<input>` No closing tag needed.

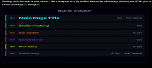
⑤ **NESTED ELEMENTS** — HTML elements can be nested inside other elements. EXAMPLE `<div> <p>Hello</p> </div>` The `<p>` element is inside the `<div>` element.

Note: Notepad me bhi code likhke chale skte hai. Save with extension .html, then open with Chrome.

Headings h1 – h6

Q/What are Headings?

Headings create hierarchy in your content — like a newspaper has a big headline, then smaller sub-headings, then body text. HTML gives us 6 levels of headings: h1 through h6



SEO Basics of Headings

Search engines and assistive technologies analyze HTML headings to understand:

- page structure
- content hierarchy
- important topics

Proper heading usage improves:

- readability
- accessibility
- SEO
- content organization

Best Practices

- Prefer one primary `<h1>` for the main page topic.
- Use headings hierarchically:`<h1>` → `<h2>` → `<h3>`
- Use headings to describe section content clearly.
- Maintain logical heading structure when possible.
- Use CSS for styling — not heading tags purely for font size.
- Include important keywords naturally in headings.
- Ensure headings improve accessibility and screen-reader navigation.

Important Interview Insight

`<h1>`

does NOT mean: "make text bigger"

It means: "this is the most important heading of this section/page."

That semantic meaning is extremely important in:

SEO
accessibility
screen readers
document structure

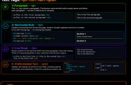
Common Mistakes

- Using headings only for visual styling.
- Randomly skipping heading hierarchy without structural reason.
- Using vague headings like:

"Title"
"Section"
"Content"

- Using multiple unrelated `<h1>` tags without clear structure.
- Using heading tags where semantic meaning is unnecessary.

Text Tags: <p> <hr>
 <pre>



1) Paragraph — <p>

The `<p>` tag creates a paragraph. The browser automatically adds margin above and below each paragraph — we don't need to do it manually.

```
<p>This is the first paragraph.</p>
<p>This is the second paragraph.</p>
```

This is the first paragraph.
This is the second paragraph.

2) Horizontal Rule — <hr>

Creates a horizontal line across the page. Used to visually separate sections of content. It is a self-closing tag — no closing tag needed.

```
<h2>Section 1</h2>
<p>Some content here.</p>
<hr> <!-- Self-closing! No </hr> needed -->
<h2>Section 2</h2>
```

Section 1
Some content here.
Section 2

**3) Line Break —
**

Moves content to the next line within the same paragraph. Also self-closing. Think of it as pressing Enter in your code but in the browser output.

```
<p>This is the first line.<br>This is the second line.<br>
This is the third line.</p>
```

This is the first line.
This is the second line.
This is the third line.

4) Preformatted Text — <pre>

Displays text exactly as written in the HTML, including spaces, line breaks, and indentation. Useful for code blocks, poems, etc.

```
<pre>
Line 1
  Line 2 with  spaces
Line 3
    Line 4
</pre>
```

```
Line 1
    Line 2 with  spaces
Line 3
    Line 4
```

Comments in HTML

Comments are notes for us (the developer) —
the browser completely ignores them. They never show up on the webpage.
Perfect for explaining code or temporarily disabling markup.

```
<!-- This is a comment -->
```

Note: HTML comments can span multiple lines. They are useful for documentation and debugging, and can be used to temporarily disable code without deleting it.

```
<!--
  This is a
  multi-line
  comment.
-->
```

Accessibility & Semantics

Q) What is Accessibility?
Accessibility means making your website usable by EVERYONE — including people with disabilities, such as visual impairments who use screen readers. Screen readers are assistive technologies that read HTML content aloud. Proper HTML structure (especially headings, landmarks, labels, etc.) helps screen readers understand content correctly.

Enhanced Accessibility
Headings improve navigation for users with disabilities. Screen readers rely on headings to navigate content efficiently.

Consistency Across Devices
A hierarchical heading structure ensures content remains logical and easy to navigate across devices — mobile, desktop, and assistive technologies. It works best when the document's semantic structure is strong.

Semantic Structure

Q) What is Semantic HTML?
Semantic HTML means using the right tag for the right content. Don't just use `<div>` for everything. Use tags that describe their meaning.

`<header>` for header · `<nav>` for navigation · `<main>` for main content · `<article>` for articles · `<footer>` for footer

Non-Semantic

```
<div class="header"> ... </div>
<div class="nav"> ... </div>
```

Semantic

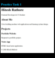
```
<header> ... </header>
<nav> ... </nav>
```

Interview Insight

- Semantic HTML helps SEO — search engines understand your content better.
- Semantic HTML helps accessibility — screen readers work correctly.
- Semantic HTML makes your code maintainable — easier for other developers to read.
- Using `<h1>` in footer is wrong — lower hierarchy tags should be used there.

Now it's time to implement these concepts practically.

Practice Task 1



Hitesh Rathore
Frontend Developer & CS Student
About Me
I love building modern web applications and learning system design.
Projects
Portfolio Website
Responsive portfolio project.
Notes App
HTML based notes application.
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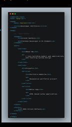
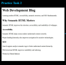
Code

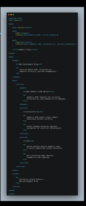


Practice Task 2

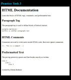


Web Development Blog
Learning modern HTML, accessibility, semantic structure, and SEO fundamentals.
Why Semantic HTML Matters
Semantic HTML improves the structure, accessibility, and readability of webpages.
Accessibility
Semantic HTML helps screen readers understand content correctly.
Proper heading hierarchy also improves navigation for assistive technologies.
SEO
Search engines analyze semantic tags to better understand content hierarchy.
Well-structured HTML improves readability and indexing.
Written by Hitesh Rathore

Code

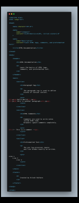


Practice Task 3

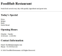


HTML Documentation
Learn the basics of HTML tags, comments, and preformatted text.
Paragraph Tag
The paragraph tag is used to define blocks of textual content.
HTML Comments
Comments are used to write notes inside HTML code. Browsers ignore comments.
Preformatted Text
The pre tag preserves spaces and line breaks exactly as written.
Created by Hitesh Rathore

Code

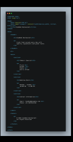


Practice Task 4

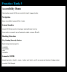


FoodHub Restaurant
Fresh food served every day with quality ingredients and great taste.
Today's Special
Pizza
Burger
Pasta
Garlic Bread
Opening Hours
Monday - Sunday
10:00 AM - 11:00 PM
Contact Information
Email: foodhub@example.com
© 2026 FoodHub Restaurant. All Rights Reserved.

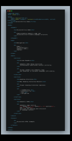
Code



Practice Task 5



Accessibility Demo
Navigation
Screen Readers
Heading Structure
Why Heading Hierarchy Matters
Semantic HTML

Code